Homework 7: Hair & DNA Evidence

Paper track. Pairs with Home Lab 7 (only if you have a magnifier/microscope).

Part A: Medullary index (2 pts each)

Compute the MI and say human or animal (human $< 1/3 \approx 0.33$; animal $> 1/2$).

1. Medulla 10 μm, shaft 50 μm.
2. Medulla 22 μm, shaft 40 μm.
3. Medulla 18 μm, shaft 60 μm.
4. Medulla 30 μm, shaft 45 μm.

Part B: Growth cycle (1 pt each)

1. List the four phases in order.
2. Which phase are most scalp hairs in? How long is it?
3. Which phase is the "resting" phase, and how long does it last?
4. A hair at a scene has a hard, dry, club-shaped root and no tissue. Which phase did it come from? Is it good for nuclear DNA?
5. About how many inches does human hair grow per year?

Part C: Classic questions (2 pts each)

1. "The hair on a corpse keeps growing." True or false — explain.
2. Hair's absorbent property makes it important in what kind of poisoning, and why?
3. Why is a forcibly pulled hair more useful for DNA than a naturally shed one?

Part D: Reading a gel (4 pts)

A DNA gel has a "Crime Scene" lane and lanes for Suspects 1–4.

- Crime Scene bands (distance from wells, in cm): 1.0, 2.5, 4.0, 5.5
- Suspect 1: 1.0, 3.0, 4.0, 6.0
- Suspect 2: 1.0, 2.5, 4.0, 5.5
- Suspect 3: 1.5, 2.5, 4.5, 5.5
- Suspect 4: 1.0, 2.5, 3.5, 5.0

1. Which suspect matches the crime scene? Explain how you decided.
2. Which direction did the smallest fragments travel, and why?
3. Name two kinds of crime-scene evidence that could have provided this DNA.

Answer Key

Part A: 1. 0.20 → human. 2. 0.55 → animal. 3. 0.30 → human. 4. 0.67 → animal.

Part B:

1. Anagen, catagen, telogen, exogen.
2. Anagen; 2–7 years.
3. Telogen; about 2–4 months.
4. Telogen (naturally shed); poor for nuclear DNA — mitochondrial only.
5. About 6 inches.

Part C:

1. False — after death the skin dehydrates and retracts, pulling back from the hair bases so the hairs *look* longer. No new growth.
2. Arsenic (and other heavy-metal) poisoning — arsenic binds to keratin and stays fixed in place as the hair grows, so segments along the hair preserve a datable timeline of exposure.
3. A pulled hair carries the follicle and root-sheath tissue, which is rich in nuclear DNA; a shed hair is just the dead shaft (mitochondrial DNA only).

Part D:

1. **Suspect 2** — every band (1.0, 2.5, 4.0, 5.5) lines up with the crime-scene lane.
2. Toward the positive electrode — DNA is negatively charged, and smaller fragments move through the gel faster, so they travel farther.
3. Any two: blood, saliva, semen, skin/touch cells, hair with the root attached, bone, teeth.